

Module 2: Routing Concepts

INTRODUCTION

Think of an instance where you wish to send a letter to a distant friend. Instead of driving there, you simply put your letter in a postbox. Then, the post office will study the address provided in the envelope, select the appropriate route (by truck, by air, or rail), and continue moving it through numerous sorting centers till it reaches its intended destination.

Routing refers to the process used to determine the best possible route to transport data from one point to another in a network. The entire routing process depends on the use of specific hardware equipment known as routers. In essence, once the data reaches the router, the destination IP address is examined, after which the router looks into its own "directory" (routing table).

Types of Routing

Before looking at the pros and cons, it helps to understand that routing generally falls into two main categories:

1. **Static Routing:** A network administrator manually configures and maps out the paths in the routing table. The routes never change unless a human changes them.
2. **Dynamic Routing:** Routers use complex mathematical formulas called routing protocols (like OSPF or BGP) to talk to each other, discover networks automatically, and dynamically choose the best paths based on real-time network conditions.

Advantages and Disadvantages

1. Static Routing

- **Advantage Static Routing** No CPU Overhead: Because the routes are hardcoded, the router doesn't have to spend processing power or RAM calculating paths.
- **Predictable Traffic:** Outbound data always takes the exact same path, making traffic behavior highly predictable.
- **Higher Security:** No routing protocols are actively broadcasting network maps across the wires, reducing the risk of interception.

Disadvantage Static Routing

- **Not Scalable:** Manually configuring every single route is nearly impossible in large or growing networks.
- **No Redundancy / Fault Tolerance:** If a link goes down, static routing cannot automatically reroute traffic. The connection simply breaks until an administrator manually fixes it.
- **High Maintenance:** Every time a new network segment is added or changed, an administrator must log into the routers to update them

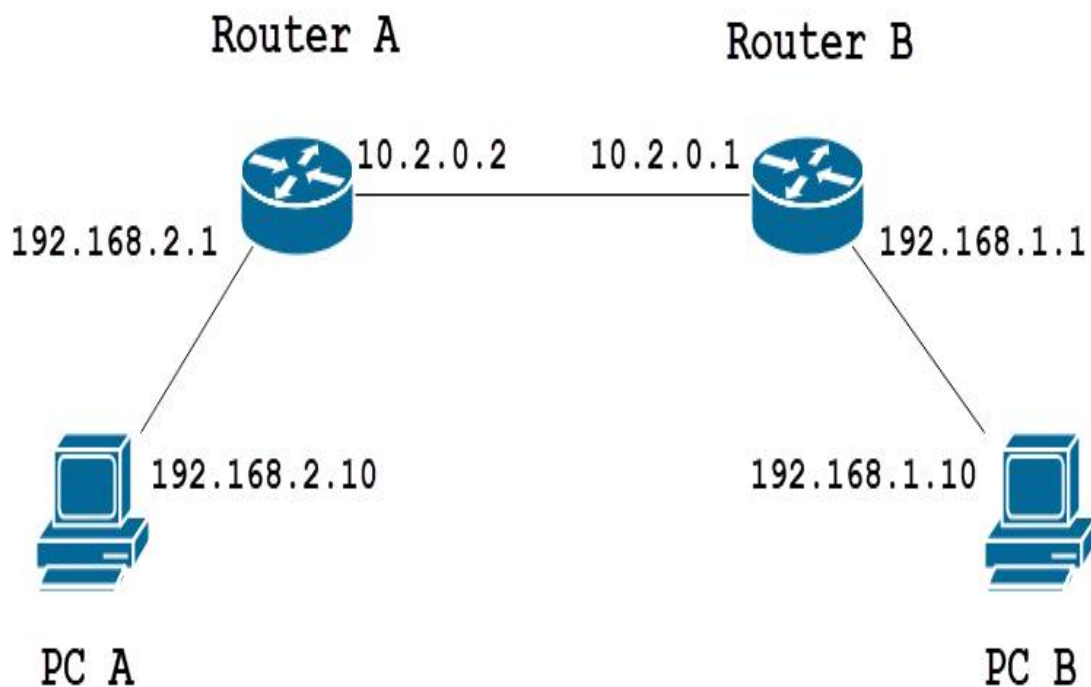
2. Dynamic Routing

Advantage of Dynamic Routing

- High Scalability: Routers automatically discover new networks and share them with peer routers. You can add hundreds of devices without manual reconfiguration.
- Automatic Rerouting (Fault Tolerance): If a specific fiber cable gets cut or a router fails, dynamic protocols instantly detect the failure, calculate a backup path, and keep traffic flowing.
- Simpler Management: Once the initial configuration is complete, the network largely manages its own topology maps.

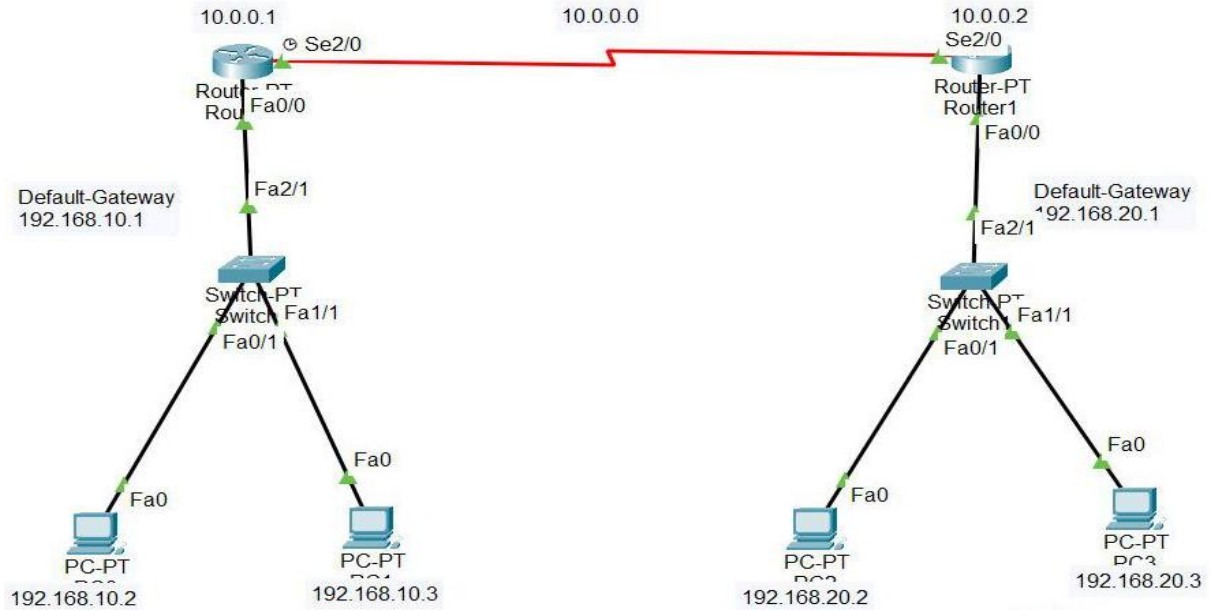
Example: (Routing Concept)

Using Cisco Packet Tracer. Configure a functional routing path between two separate Local Area Networks (LANs) using Cisco routers, and verify the resulting routing tables.



Activity no. 2

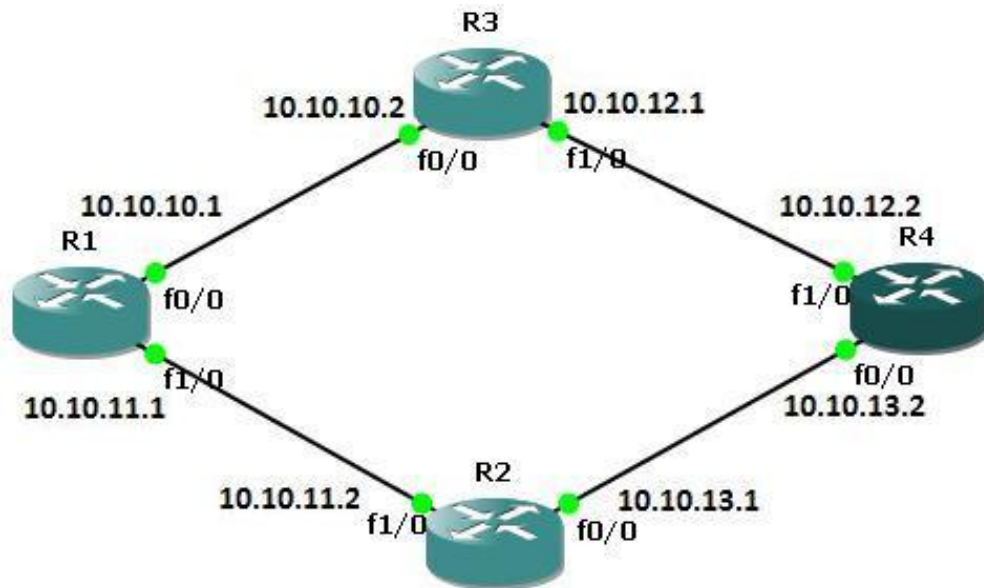
A. Using Cisco Packet Tracer. Configure a functional routing path between two separate Local Area Networks (LANs) using Cisco routers. Write your observation to the box provided.



Instructor's Signature : _____

Date: _____

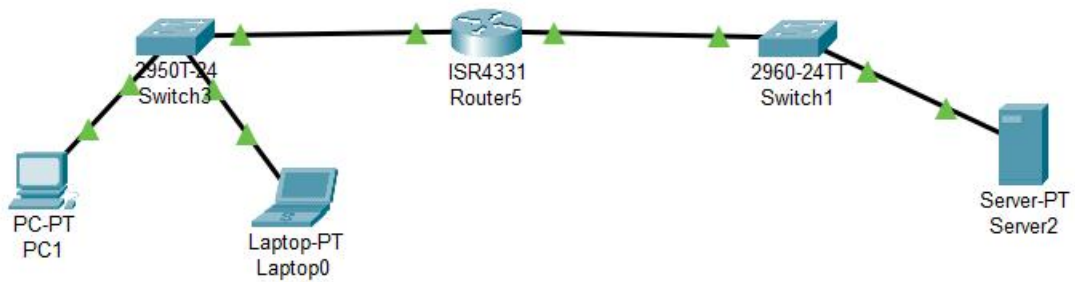
B. Using Cisco Packet Tracer. Configure a functional routing path between Cisco routers, and verify the connectivity using ping command. Write your observation on the box provided.



Instructor's Signature : _____

Date: _____

C. Using Cisco Packet Tracer. Configure a functional routing path between Cisco routers, and verify the connectivity using ping command between PC1 and Server 2. Write your observation on the box provided.



Instructor's Signature : _____

Date: _____

D. What are the difference between Static routing and Dynamic Routing? Write your answer to the box provided.

Instructor's Signature : _____

Date: _____